

One-class hybrid SVM-CAE based MEWMA monitoring scheme

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Abstract

This report develops and evaluates a hybrid multivariate monitoring scheme that combines a one-dimensional convolutional autoencoder (CAE), a one-class support vector machine (OC-SVM) and a multivariate exponentially weighted moving average (MEWMA) control chart. For every window of 32 observations, the CAE's reconstruction error and the OC-SVM's anomaly score on the CAE features form a two-dimensional monitoring vector that the MEWMA accumulates over time. The scheme is studied on a simulated nonlinear process driven by a latent vector autoregression, using five independent datasets for representation learning, detector fitting, empirical calibration of the control limit to an in-control average run length (ARL) of 370, independent in-control validation, and out-of-control evaluation under sustained latent mean shifts of magnitude 0.2 to 3.0. On independent in-control data the hybrid chart was somewhat more liberal, and a classical raw-data MEWMA somewhat more conservative, than the design target. Out of control, the hybrid chart had a shorter average run length than both of its single-component ablations at every shift, but it did not outperform the classical MEWMA: the two were indistinguishable at the smallest shift, and the classical chart detected faster at every shift of 0.4 or more. Post-hoc diagnostics attribute this mainly to the weak response of distance-type learned features to a directional mean shift, amplified at large shifts by the startup behaviour of the charts.

Declaration

I, *Stephen James Pienaar*, declare that this research report, submitted in partial fulfillment of the degree *BCom(Hons) Statistics and Data Science*, at the University of Pretoria, is my own work and has not been previously submitted at this or any other tertiary institution.

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1 Introduction

1.1 General concepts in SPC and ML

1.1.1 Statistical Process Control

Statistical process control (SPC) is a collection of statistical techniques for monitoring process stability and improving capability by reducing variability [15]. Its principal tool is the control chart, which plots a monitoring statistic against a control limit; a value beyond the limit signals a possible out-of-control (OOC) departure from the in-control (IC) state. Charts are judged by their run lengths, usually through the average run length (ARL): a long in-control ARL (ARL_0) means few false alarms and a short out-of-control ARL (ARL_1) fast detection [20]. Memoryless Shewhart-type charts suit large shifts, whereas memory-type charts such as the cumulative sum (CUSUM) [17] and the exponentially weighted moving average (EWMA) [21] accumulate evidence and suit small, persistent shifts; the multivariate EWMA (MEWMA) [11] extends this to correlated characteristics. These charts assume independent, normally distributed observations, and when either assumption fails the realised ARL_0 can differ substantially from its nominal value [11, 20].

1.1.2 Machine Learning

Machine learning (ML) offers data-driven alternatives that need no parametric process model; because labelled faults are rare, the relevant methods model normal behaviour only [18]. An autoencoder (AE) is a neural network trained to reconstruct its input through a low-dimensional bottleneck [6]; trained on normal data, its reconstruction error measures departure from normal structure. A convolutional autoencoder (CAE) uses convolutional layers that extract local patterns from inputs such as short multivariate time-series windows [8, 14]. The one-class support vector machine (OC-SVM), an unsupervised extension of the support vector machine [4], estimates a region containing most of the normal data and scores new observations relative to its boundary [22]. On CAE features the two signals are complementary: the score asks whether a window's representation looks typical, the reconstruction error whether the network can reproduce the window at all.

1.2 Problem statement and significance of the study

Nonlinear relationships, serial dependence and non-normality weaken the distributional basis of standard control limits [11, 20]. Learned detectors can represent such structure, but they are usually applied observation by observation against a fixed threshold, without accumulating evidence or a run-length interpretation of their alarm rate [3, 5, 12]. This study places CAE reconstruction information and OC-SVM anomaly-score information inside a memory-type MEWMA chart and evaluates the result as a

control chart: its limit is calibrated to a nominal ARL_0 and its run lengths are compared with those of a classical MEWMA calibrated to the same target, so any benefit of learning is judged on the classical chart's own terms.

1.3 Research aim and objectives

The aim is to develop and evaluate a hybrid ML-based MEWMA monitoring scheme that integrates a CAE and an OC-SVM for detecting process shifts in multivariate data. The objectives are to:

- construct a CAE and OC-SVM hybrid that combines both signals in one monitoring vector;
- investigate its run-length characteristics, with particular interest in small shifts;
- evaluate the CAE and OC-SVM components using appropriate diagnostics;
- compare the hybrid with a classical MEWMA under a common in-control design target;
- examine a range of shift magnitudes and, in a limited analysis, the smoothing constant; and
- demonstrate the scheme on simulated multivariate process data.

1.4 Research outline

The study is confined to unsupervised monitoring of a simulated nonlinear process, so the true state is always known. It covers Phase I, in which all components and limits are estimated from in-control data, and Phase II, in which the frozen scheme is applied to independent sequences [20]. Sections 2–6 present the literature review, methodology, results, practical application and conclusions.

2 Literature Review

Memory-type charting began with the CUSUM of Page [17] and the geometric moving average of Roberts [21]. For several correlated characteristics, two traditions developed. The first extends univariate charts through Hotelling's T^2 statistic, culminating for the memory-type case in the MEWMA chart, whose limit is set to achieve a specified ARL_0 and whose ARL depends on a mean shift only through its Mahalanobis distance [11]. The second, reviewed by MacGregor and Kourti [13], projects the observations onto a few latent variables and monitors two complementary statistics: T^2 on the retained scores and a squared prediction error (SPE) on the residual. Both traditions rest on linear structure and on independence and normality assumptions whose failure can distort run-length behaviour [20]; distribution-free schemes relax the parametric assumptions [2].

Representation learning offers a nonlinear counterpart to the projection step. Autoencoders learn low-dimensional codes that outperform principal components for dimension reduction [6], convolutional autoencoders add locality and weight sharing [14], and one-dimensional convolutional networks suit sensor signals when labelled data are scarce [7]. The analogy with latent-variable monitoring is direct: the autoencoder plays the role of the principal component model and its reconstruction error that of the SPE. For the latent space, one-class methods model normality when abnormal examples are too scarce to model [18], and the OC-SVM scores observations relative to an estimated support boundary [22]. Feeding learned latent codes to an OC-SVM is established practice [1, 12, 19], and Guzman and Howe [5] report that combining autoencoder reconstruction error with an OC-SVM beats either component alone; all of these detectors, however, classify each observation against a fixed threshold.

Hybrid ML and SPC research has followed two strands. In the first, a learner operates on charting statistics while the chart remains classical: convolutional and recurrent networks recognise control-chart patterns [23], deep networks and support vector regression model EWMA-type statistics for censored reliability data [9, 10], and models trained on classical charting statistics widen the range of detectable shifts [16]. These learners need labelled out-of-control examples or a parametric model from which training data are simulated. In the second strand, representation learning enters the monitoring statistic itself: Biegel et al. [3] compute T^2 and SPE statistics from a deep autoencoder’s latent representation and reconstruction error on a real sheet-metal forming process. That approach is unsupervised and acts on process observations, but its statistics are memoryless and it is assessed by classification metrics rather than run lengths. Within the literature reviewed, no study combines a CAE reconstruction error with an OC-SVM score on the CAE representation in a memory-type chart whose limit is calibrated to a nominal ARL_0 and compared with a classical MEWMA under the same target; this study addresses that gap.

3 Methodology

3.1 Simulated process and study design

A latent state $z_t \in \mathbb{R}^8$ follows a stable mean-adjusted VAR(1), and the four observed variables are a nonlinear function of it:

$$z_t = \mu_t + \Phi(z_{t-1} - \mu_{t-1}) + \eta_t, \quad \eta_t \sim N(0, 0.2I_8); \quad x_t = W_2 \tanh(z_t) + \varepsilon_t, \quad \varepsilon_t \sim N(0, 0.05I_4). \quad (1)$$

Φ is symmetric tridiagonal (diagonal 0.55, adjacent off-diagonals 0.15; spectral radius ≈ 0.832), $\tanh$ acts componentwise and W_2 is a fixed 4×8 mixing matrix (Appendix A). Each trajectory starts at $z_0 = 0$ and discards 500 burn-in steps. Because the map is nonlinear and $p = 4 < 8$, no linear projection of x_t recovers z_t ; the latent state is never supplied to any model. The fault is a sustained latent mean shift

Hybrid chart H (ablations: S uses s_t only, R uses ℓ_t only)

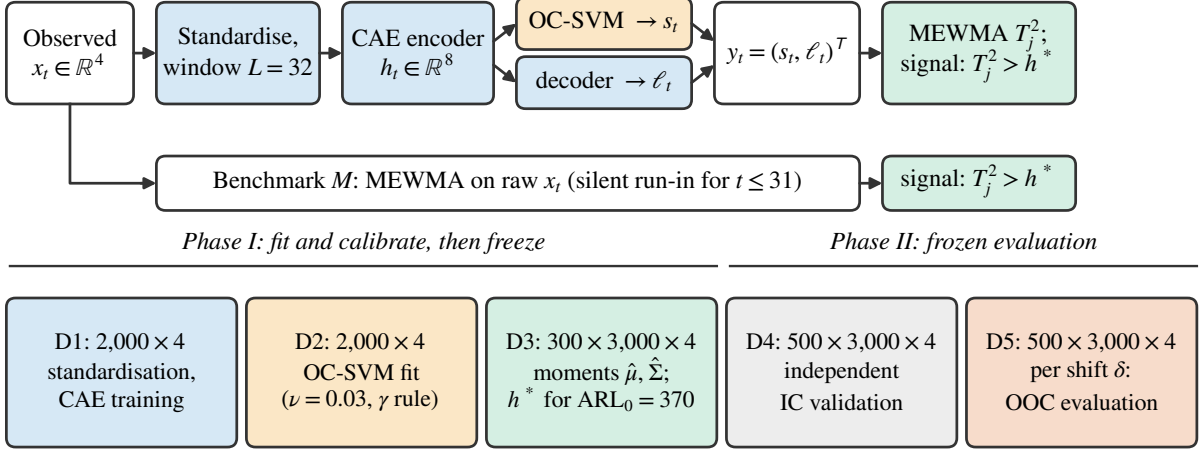


Figure 1: Hybrid and benchmark monitoring paths (top) and the roles of the five datasets (bottom). Box shading marks the dataset on which a stage is fitted or calibrated; every chart first reports at raw time $t = 32$ and has $J = 2969$ opportunities per trajectory.

$\mu_t = \delta v$ in the fixed direction $v = (1, 1, 0, \dots, 0)^T / \sqrt{2}$, starting at the first retained observation ($\tau = 1$), so δ is the Euclidean size of the shift and every out-of-control trajectory is post-change from its first observation. Fifteen magnitudes, $\delta = 0.2, 0.4, \dots, 3.0$, are studied; no other fault type is used.

Five datasets generated from independent random streams have separate roles (Figure 1): D1–D3 fit and calibrate the scheme in Phase I, after which everything is frozen and applied unchanged to D4 (independent in-control validation) and D5 (out-of-control evaluation) in Phase II.

3.2 Learned components

Windows. D1 is split chronologically into training rows $[0, 1600)$ and validation rows $[1600, 2000)$ before any window is formed. Each variable is standardised with the training-segment mean and standard deviation, which are then frozen. Windows $X_t \in \mathbb{R}^{4 \times 32}$ of the $L = 32$ most recent standardised observations are formed with stride 1 inside a segment or trajectory, so a 3000-observation trajectory yields $J = 2969$ windows, the first ending at $t = 32$.

CAE. The encoder applies two one-dimensional convolutions ($4 \rightarrow 8 \rightarrow 16$ channels, kernel 3, stride 2, ReLU), which halve the window length twice ($32 \rightarrow 16 \rightarrow 8$), and a linear bottleneck producing $h_t \in \mathbb{R}^8$; the decoder mirrors this with a linear layer and two transposed convolutions (3180 parameters; Appendix A). The network was trained on the 1569 D1 training windows to minimise the mean squared reconstruction error (Adam, learning rate 10^{-3} , batch size 64, at most 100 epochs, early-stopping patience 10), and the checkpoint with the lowest validation loss (epoch 33) was frozen. The reconstruction

component is the summed squared prediction error of the standardised window and its floored logarithm,

$$\text{SPE}_t = \sum_{c=1}^4 \sum_{k=1}^{32} (X_{t,ck} - \hat{X}_{t,ck})^2, \quad \ell_t = \log \max(\text{SPE}_t, 10^{-12}). \quad (2)$$

OC-SVM. The 1969 D2 windows are passed through the frozen standardisation and encoder, and an RBF OC-SVM is fitted to the resulting features directly, without further scaling:

$$k(h, h') = \exp(-\gamma \|h - h'\|^2), \quad \nu = 0.03, \quad \gamma = \left[\text{median}_{i < j} \|h_i - h_j\|^2 \right]^{-1} \approx 0.0305, \quad (3)$$

with the median taken over all $\binom{1969}{2}$ pairs. The parameter ν bounds the fraction of training windows outside the fitted boundary from above and the fraction of support vectors from below [22]; the fit has 78 support vectors. The monitored score is the negated decision function, $s_t = \rho - \sum_j \alpha_j k(h_{(j)}, h_t)$, so that a larger s_t means a more abnormal window.

3.3 Monitoring charts and calibration

The hybrid monitoring vector is $y_t = (s_t, \ell_t)^\top$. For a chart input u_j with in-control mean $\hat{\mu}$ and covariance $\hat{\Sigma}$, the MEWMA state and statistic are [11]

$$Z_j = \lambda u_j + (1 - \lambda)Z_{j-1}, \quad Z_0 = \hat{\mu}, \quad T_j^2 = (Z_j - \hat{\mu})^\top (c_j \hat{\Sigma})^{-1} (Z_j - \hat{\mu}), \quad c_j = \frac{\lambda}{2 - \lambda} [1 - (1 - \lambda)^{2j}], \quad (4)$$

where j indexes monitoring opportunities ($j = 1$ at $t = 32$) and the exact finite-start factor c_j is always used. Six configurations of four charts are evaluated (Table 1). The hybrid chart H monitors y_t at the primary $\lambda = 0.05$, with $\lambda = 0.10$ and 0.20 as a sensitivity analysis; the ablations S and R monitor s_t and ℓ_t alone, and the benchmark M is a conventional MEWMA on the raw x_t , each with $\lambda = 0.05$. $\hat{\mu}$ and $\hat{\Sigma}$ are pooled over all 300×2969 D3 monitoring vectors (S and R use the matching entries) and, for M , over the aligned raw D3 observations at $t = 32, \dots, 3000$. Because the hybrid charts cannot decide before the first window is complete, M 's state is updated silently with $x_1, \dots, x_{31}$ and it first reports at $t = 32$ with factor c_{32} , so all six charts share the same 2969 opportunities.

Consecutive windows share 31 of their 32 observations, so the monitored sequences are strongly autocorrelated (Section 4.1) and $c_j \hat{\Sigma}$ is only a reference scaling, not the covariance of Z_j ; without independence, limits derived for independent data lose their nominal ARL properties [11]. Each limit is therefore calibrated empirically on D3. For a candidate h , $\widehat{\text{ARL}}_0(h)$ is the mean over the 300 D3 trajectories of the first opportunity with $T_j^2 > h$, truncated at $J = 2969$. As this is a non-decreasing step function of h , the calibrated limit

$$h^* = \min \{h \in V : \widehat{\text{ARL}}_0(h) \geq 370\}, \quad (5)$$

where V is the set of values attained by the running maxima of T_j^2 , is found exactly by bisection. The target of 370 is the ARL_0 of a conventional three-sigma Shewhart chart [20].

3.4 Evaluation design

A run length is the first opportunity j with $T_j^2 > h^*$; if none occurs by $j = 2969$, the run length is recorded as 2969 and flagged as censored by a separate indicator, so each mean is a restricted ARL. D4 yields each chart’s independent in-control run lengths, kept distinct from its D3 calibration value, and D5 its out-of-control run lengths at the 15 shifts. Because $\tau = 1$, the first window is already post-change and a signal at opportunity j corresponds to $j + 31$ post-change observations: ARL_1 is a detection delay measured from the first permitted decision, the setting in which Lowry et al. [11] also compared multivariate charts. $H(0.05)$ and $M(0.05)$ monitor the same D5 trajectories, so the primary comparison is paired,

$$\bar{d}(\delta) = \frac{1}{500} \sum_{i=1}^{500} [RL_{H,i}(\delta) - RL_{M,i}(\delta)], \quad (6)$$

with standard error $s_d/\sqrt{500}$; a positive $\bar{d}$ means that the benchmark signalled first. The design was fixed in advance: no model, moment, smoothing constant or limit was changed after D4 or D5 was examined.

4 Results

Sections 4.1–4.4 report canonical results; Section 4.5 adds post-hoc interpretation.

4.1 Behaviour of the learned components

The CAE generalised to independent in-control data, with a mean squared reconstruction error per window element of 0.66 on the D1 validation windows and 0.67 on D3. The OC-SVM met its ν -property in sample (3.0% of D2 windows outside the boundary; 4.0% support vectors), but 6.2% of D3 windows fell outside, so its boundary was somewhat tight out of sample, which the D3-calibrated limit absorbs. Figure 2 shows that the components were weakly correlated ($r = -0.15$), that s_t was right-skewed (skewness 0.55) while ℓ_t was nearly symmetric, and that both were strongly serially dependent: their lag-one autocorrelation of 0.96 decayed almost linearly to near zero at lag 32, where windows stop sharing observations, whereas the raw variables’ lag-one autocorrelations of 0.45–0.60 fell below 0.07 by lag 10. This overlap-induced dependence is why the limits were calibrated empirically.

4.2 In-control validation

Table 1 sets each chart’s D3 calibration value beside its independent D4 performance; no D4 run length was censored (one D3 trajectory was censored when calibrating $H(0.20)$). The realised ARL_0 departed

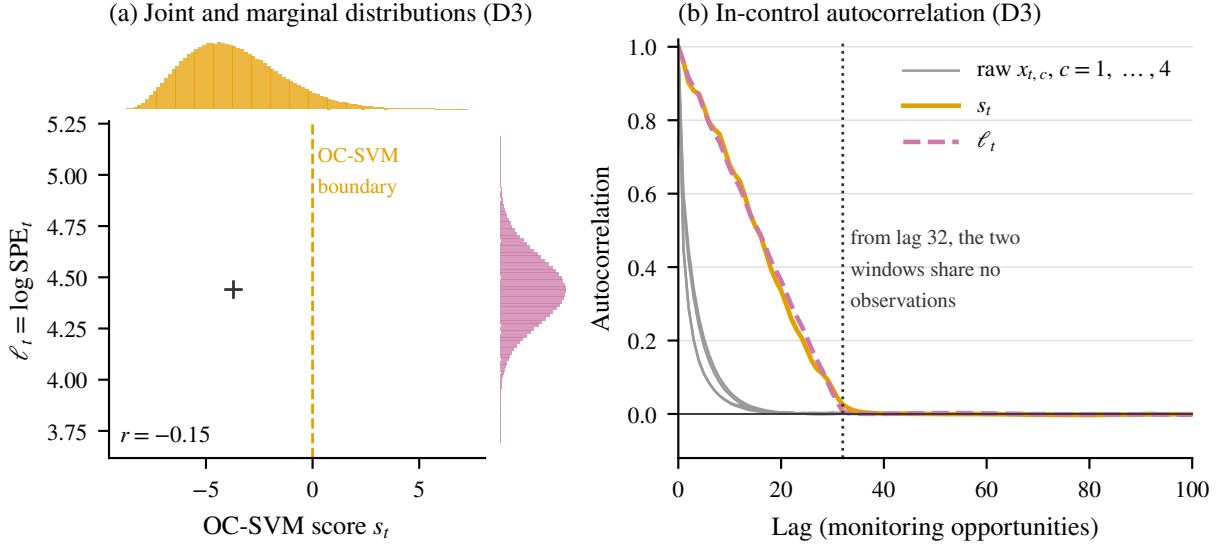


Figure 2: In-control monitoring vectors on D3 (300×2969 windows, recomputed from the frozen models; they reproduce the stored D3 moments exactly). (a) Joint distribution of (s_t, ℓ_t) (grey shading: log count; cross: $\hat{\mu}$) with marginal histograms. (b) Autocorrelation of s_t , ℓ_t and the four raw variables, pooled over trajectories.

Table 1: Calibrated limit, D3 calibration ARL_0 and independent D4 in-control performance (500 trajectories; SE in parentheses) of the six charts.

Chart	Input	h^*	D3 ARL_0	D4 ARL_0 (SE)	D4 SDRL	D4 MRL
$H(0.05)$	$y_t = (s_t, \ell_t)^\top$	111.08	371.0	336.5 (14.7)	328.8	230.5
$H(0.10)$	$y_t = (s_t, \ell_t)^\top$	83.59	370.5	322.3 (13.6)	304.2	226.0
$H(0.20)$	$y_t = (s_t, \ell_t)^\top$	51.43	372.4	315.4 (13.7)	305.9	210.5
$S(0.05)$	s_t	66.05	370.6	351.4 (14.9)	333.0	246.5
$R(0.05)$	ℓ_t	71.03	371.3	363.4 (16.1)	361.0	253.5
$M(0.05)$	raw x_t	46.52	372.4	403.5 (17.9)	401.3	268.5

from 370 in opposite directions: the hybrid configurations ran shorter (315.4–336.5, i.e. 2.3–4.0 D4 standard errors below target), the ablations stayed close (351.4 and 363.4) and $M(0.05)$ ran longer (403.5, 1.9 standard errors above); on the same D4 trajectories, $M(0.05)$ ’s run lengths exceeded those of $H(0.05)$ by 67.0 on average (paired SE 22.2). Deviations of this size can arise from sampling variability in both the finite calibration bank and D4, but their direction matters: $H(0.05)$ was the more liberal chart in control, which, if anything, favours it out of control.

4.3 Primary comparison: $H(0.05)$ versus $M(0.05)$

Figure 3(a)–(b) covers the complete shift grid; no D5 run length was censored. At $\delta = 0.2$ the charts were indistinguishable (ARL_1 282.4 and 285.6; $\bar{d} = -3.2$, paired SE 15.9). At every $\delta \geq 0.4$ the benchmark signalled first on average, by more than ten paired standard errors (Appendix C). The absolute gap peaked at $\delta = 0.4$ – 0.6 ($\bar{d} \approx 98$ – 106 opportunities) and then narrowed, while the relative gap widened: $M(0.05)$ signalled at its first opportunity in 41% of trajectories at $\delta = 1.0$ and in essentially all from

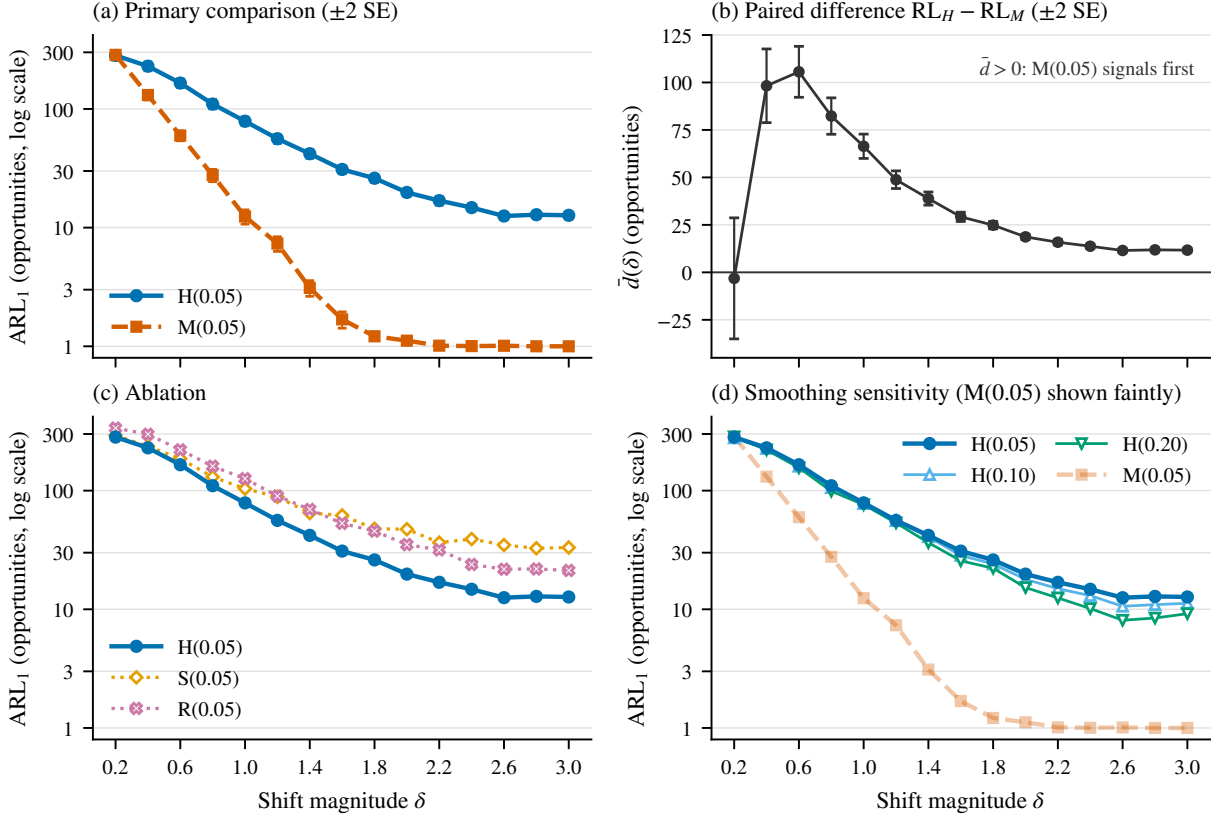


Figure 3: Out-of-control results on D5 (500 trajectories per δ ; log scale in (a), (c), (d)). (a) ARL_1 of $H(0.05)$ and $M(0.05)$; (b) paired mean difference $\bar{d}(\delta)$ of Equation (6); (c) $H(0.05)$ and its ablations; (d) the three hybrid smoothing constants, with $M(0.05)$ for reference.

$\delta = 2.2$, whereas $H(0.05)$ never did so and levelled off near an ARL_1 of 12.5. Counted in post-change observations ($j + 31$), the mean delays were 110 against 43 at $\delta = 1.0$ and 44 against 32 at $\delta = 3.0$, a ratio of 1.4–2.5 for $\delta \geq 0.4$, compared with up to 21 in monitoring opportunities. Under this sustained directional mean shift, the classical benchmark therefore detected faster at every shift from 0.4 upwards, despite being the more conservative chart in control.

4.4 Ablation and smoothing sensitivity

Figure 3(c) shows that $H(0.05)$ had a lower ARL_1 than both ablations at all 15 shifts. Paired over the same trajectories (Appendix C), its advantage over $R(0.05)$ was clear throughout (8.6–69.0 opportunities, more than three paired SE), whereas that over $S(0.05)$ was within sampling error at $\delta \leq 0.4$ (−9.7, SE 14.5; −6.4, SE 11.3) and exceeded two paired SE from $\delta = 0.6$; the score was the stronger single component up to $\delta = 1.4$ and the reconstruction error beyond it. Figure 3(d) shows that a larger λ changed little at small shifts and helped at larger ones, as expected for EWMA-type charts [11]: $H(0.20)$ had the lowest hybrid ARL_1 at every $\delta \geq 0.4$, by 2.6–11.0 opportunities relative to $H(0.05)$. Its D4 ARL_0 was also lower (paired difference −21.1, SE 11.0), so part of this gain may reflect a more liberal realised calibration, and

no smoothing constant in the pre-specified range brought the hybrid close to $M(0.05)$.

4.5 Interpretation

Two features of the design explain these results. First, the fault is a sustained mean shift in a fixed direction, the alternative that a linear, accumulating MEWMA on x_t is built to detect [11], whereas the hybrid accumulates two distance-type summaries of each window, which respond weakly to small location shifts. Second, with $\tau = 1$ the benchmark’s first report already contains 32 accumulated post-change updates, whereas the hybrid’s first statistic is the Mahalanobis distance of a single window, compared with a limit set by the strongly persistent steady state. Post-hoc diagnostics, which are not part of the pre-specified experiment, support both explanations: the component means moved by only 0.10–0.21 in-control standard deviations at $\delta = 0.4$ (Appendix B), and a separate diagnostic audit estimated that y_t retains 2–36% of the raw data’s mean-shift information per update, that startup mechanics account for most of the gap in monitoring opportunities at $\delta \geq 1.8$, and that with the change after 100 in-control reports $H(0.05)$ still needed 2.0–2.5 times as many post-change observations as $M(0.05)$ for $\delta \geq 0.6$. These diagnostics explain the canonical result; they do not change it.

5 Application

The scheme was evaluated only on simulated data; no industrial or benchmark process, such as the Tennessee Eastman Process, was monitored in this study. In practice it would be deployed in the same two phases. In Phase I, a stable reference period of normal operation is divided chronologically into three parts that mirror D1–D3: standardisation and CAE training, OC-SVM fitting on the frozen encoder’s features, and estimation of the moments of y_t with empirical calibration of h^* to the in-control ARL the plant can tolerate. In Phase II, each new observation completes a window of the 32 most recent standardised observations; one CAE pass and one OC-SVM evaluation give s_t and ℓ_t , which update the MEWMA, and a signal prompts investigation rather than automatic correction. The computation is light: the CAE has 3180 parameters and the OC-SVM 78 support vectors, and the complete production run of this study (data generation, training, calibration and Phase II evaluation of all six charts) took 207 seconds on a single laptop processor thread.

The practical demands are nonetheless real. The reference period must be genuinely in control, because faults present in it would be learned as normal. The persistent window features make the first reports after a restart conservative: no in-control D4 trajectory of $H(0.05)$ signalled before its eleventh report, whereas $M(0.05)$ could signal at its first. Legitimate process changes require refitting and recalibration on fresh in-control data, and a signal is harder to trace to individual variables than with a raw-data MEWMA, whose state vector indicates the direction of a shift [11]. Above all, the evidence concerns one simulated

process and one fault family, for which the raw-data benchmark was faster; external validation on the Tennessee Eastman Process or real plant data is needed before industrial use can be recommended.

6 Conclusion

6.1 Summary

This study developed a hybrid monitoring scheme in which a CAE supplies a reconstruction error and learned features, an OC-SVM scores those features, and a MEWMA chart accumulates the vector (s_t, ℓ_t) . It was implemented on a simulated nonlinear process with five independent datasets, and every chart was calibrated empirically to an in-control ARL of 370. The learned components behaved largely as intended, and on independent in-control data the hybrid was somewhat more liberal, and the classical benchmark somewhat more conservative, than the target. The hybrid had a lower ARL_1 than both single-component ablations at every shift (within sampling error only against $S(0.05)$ at $\delta \leq 0.4$), so the combined vector carries more usable information than either learned signal alone, but it did not outperform the classical MEWMA: the two were indistinguishable at $\delta = 0.2$, and the benchmark was faster at every larger shift, by up to 106 monitoring opportunities on average, with the hybrid needing 1.4 to 2.5 times as many post-change observations. Larger smoothing constants shortened the hybrid’s run lengths modestly, most in relative terms at large shifts, without closing the gap.

6.2 Recommendations

First, for sustained mean shifts in a moderately nonlinear process, a well-calibrated classical MEWMA on the raw variables should remain the default, and a learned monitor should be adopted only where it improves run-length performance against such a benchmark under a common in-control target. Second, learned reconstruction and one-class information should be monitored jointly rather than separately, since the joint chart dominated both ablations. Third, charts built on overlapping windows should be calibrated empirically on the dependent process, with delays reported both in monitoring opportunities and in post-change observations.

6.3 Future work

The hybrid should be evaluated on faults that alter the shape, variability or dependence of the data (variance changes, sensor faults, drifts and transients), for which distance-type features may be better suited than for a mean shift, and on larger, more strongly nonlinear processes. Direction-preserving learned features, dependence-aware finite-start scaling for overlapping windows, changes that occur during monitoring, and validation on the Tennessee Eastman Process and real plant data also deserve study.

References

- [1] Alaverdyan, Z., Jung, J., Bouet, R., and Lartizien, C. (2020). Regularized siamese neural network for unsupervised outlier detection on brain multiparametric magnetic resonance imaging: Application to epilepsy lesion screening. *Medical Image Analysis*, 60:101618.
- [2] Arslan, M., Yeganeh, A., Sogandi, F., Malela-Majika, J., Figueiredo, F. O., and Iqbal, A. (2026). A Rank-Based CUSUM Monitoring Scheme for Time Between Events and Amplitude Data. *Quality and Reliability Engineering International*. doi:10.1002/qre.70367.
- [3] Biegel, T., Jourdan, N., Hernandez, C., Cviko, A., and Metternich, J. (2022). Deep learning for multivariate statistical in-process control in discrete manufacturing: A case study in a sheet metal forming process. *Procedia CIRP*, 107:422–427.
- [4] Cortes, C. and Vapnik, V. (1995). Support-vector networks. *Machine Learning*, 20(3):273–297.
- [5] Guzman, E. R. and Howe, R. D. (2026). Sidewalk Hazard Detection Using a Variational Autoencoder and One-Class SVM. *Sensors*, 26(3):769.
- [6] Hinton, G. E. and Salakhutdinov, R. R. (2006). Reducing the Dimensionality of Data with Neural Networks. *Science*, 313(5786):504–507.
- [7] Kiranyaz, S., Avci, O., Abdeljaber, O., Ince, T., Gabbouj, M., and Inman, D. J. (2021). 1D convolutional neural networks and applications: A survey. *Mechanical Systems and Signal Processing*, 151:107398.
- [8] LeCun, Y., Bottou, L., Bengio, Y., and Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324.
- [9] Lee, P.-H. and Liao, S.-L. (2023). Residual Control Chart Based on a Convolutional Neural Network and Support Vector Regression for Type-I Censored Data with the Weibull Model. *Mathematics*, 12(1):74.
- [10] Lee, P.-H. and Liao, S.-L. (2024). Monitoring gamma type-I censored data using an exponentially weighted moving average control chart based on deep learning networks. *Scientific Reports*, 14(1):6458.
- [11] Lowry, C. A., Woodall, W. H., Champ, C. W., and Rigdon, S. E. (1992). A Multivariate Exponentially Weighted Moving Average Control Chart. *Technometrics*, 34(1):46–53.
- [12] Mabu, S., Fujita, K., and Kuremoto, T. (2019). Disaster Area Detection from Synthetic Aperture Radar Images Using Convolutional Autoencoder and One-class SVM. *Journal of Robotics, Networking and Artificial Life*, 6(1):48.

- [13] MacGregor, J. F. and Kourti, T. (1995). Statistical process control of multivariate processes. *Control Engineering Practice*, 3(3):403–414.
- [14] Masci, J., Meier, U., Cireşan, D., and Schmidhuber, J. (2011). Stacked Convolutional Auto-Encoders for Hierarchical Feature Extraction. In Honkela, T., Duch, W., Girolami, M., and Kaski, S., editors, *Artificial Neural Networks and Machine Learning – ICANN 2011*, volume 6791 of *Lecture Notes in Computer Science*, pages 52–59. Springer, Berlin.
- [15] Montgomery, D. C. (2012). *Statistical Quality Control*. Wiley, 7th edition.
- [16] Mukhtiar, F., Zaman, B., and Butt, N. R. (2025). Enhanced process monitoring using machine learning-based control charts for poisson-distributed data. *Engineering Applications of Artificial Intelligence*, 157:111227.
- [17] Page, E. S. (1954). Continuous inspection schemes. *Biometrika*, 41(1-2):100–115.
- [18] Pimentel, M. A., Clifton, D. A., Clifton, L., and Tarassenko, L. (2014). A review of novelty detection. *Signal Processing*, 99:215–249.
- [19] Pinon, N., Trombetta, R., and Lartizien, C. (2023). One-Class SVM on siamese neural network latent space for Unsupervised Anomaly Detection on brain MRI White Matter Hyperintensities. arXiv preprint arXiv:2304.08058.
- [20] Qiu, P. (2017). Statistical Process Control Charts as a Tool for Analyzing Big Data. In Ahmed, S. E., editor, *Big and Complex Data Analysis*, Contributions to Statistics, pages 123–138. Springer, Cham.
- [21] Roberts, S. W. (1959). Control Chart Tests Based on Geometric Moving Averages. *Technometrics*, 1(3):239–250.
- [22] Schölkopf, B., Platt, J. C., Shawe-Taylor, J., Smola, A. J., and Williamson, R. C. (2001). Estimating the Support of a High-Dimensional Distribution. *Neural Computation*, 13(7):1443–1471.
- [23] Yu, Y. and Zhang, M. (2021). Control chart recognition based on the parallel model of CNN and LSTM with GA optimization. *Expert Systems with Applications*, 185:115689.

A Simulation and model details

Table 2 gives the fixed mixing matrix of Equation (1) (full row rank; the weights of each observed variable have unit norm) and Table 3 the layers of the convolutional autoencoder, both as implemented in the authoritative repository.

Table 2: The fixed 4×8 mixing matrix W_2 , shown transposed: row k holds the weights of $\tanh(z_{t,k})$ in the four observed variables $x_{t,1}, \dots, x_{t,4}$.

k	$x_{t,1}$	$x_{t,2}$	$x_{t,3}$	$x_{t,4}$
1	-0.60447198	-0.04842081	-0.05518327	-0.04100004
2	-0.06150649	0.36085142	0.45874929	-0.84305316
3	-0.37179578	-0.83557846	-0.21939336	-0.40588165
4	-0.01161746	-0.11135410	-0.17469105	-0.11509976
5	0.05885824	-0.18152197	-0.55464952	0.26627250
6	0.52361117	0.30422580	0.30578469	0.07542546
7	0.45120458	0.15832824	-0.13652085	0.16952036
8	-0.10604239	0.07924137	-0.53672635	-0.06530930

Table 3: Layers of the convolutional autoencoder (kernel 3, stride 2 and padding 1 for every convolution; output padding 1 for the transposed convolutions).

Stage	Layer	Output shape	Parameters
Input	standardised window X_t	4×32	—
Encoder	Conv1d $4 \rightarrow 8$, ReLU	8×16	104
	Conv1d $8 \rightarrow 16$, ReLU	16×8	400
	flatten, Linear $128 \rightarrow 8$ (bottleneck h_t)	8	1032
Decoder	Linear $8 \rightarrow 128$, ReLU; unflatten	16×8	1152
	ConvTranspose1d $16 \rightarrow 8$, ReLU	8×16	392
	ConvTranspose1d $8 \rightarrow 4$ (reconstruction $\hat{X}_t$)	4×32	100
Total			3180

B Supplementary component diagnostics

Figure 4 shows a typical in-control window and its frozen-CAE reconstruction: the decoder reproduces the smooth level structure of each channel but not the high-frequency variation, which therefore dominates SPE_t . Figure 5 compares the in-control distributions of the standardised components with their distributions under three shifts. It is a post-hoc descriptive diagnostic computed from the first 100 D5 trajectories at each shift with the frozen models; it was not a pre-specified output of the experiment.

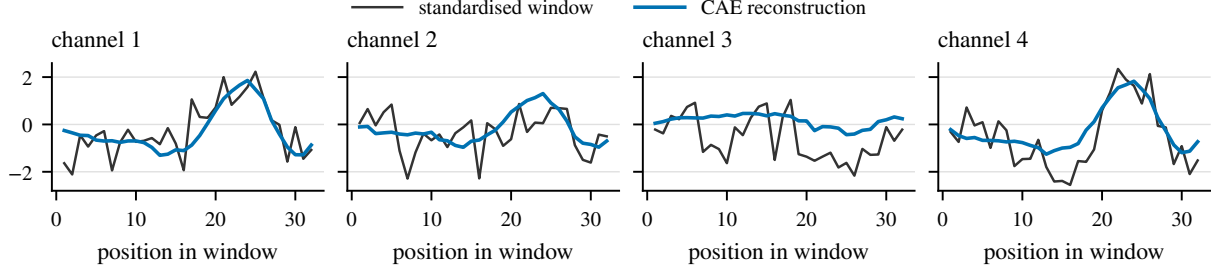


Figure 4: One in-control D3 window (trajectory 1, window 1001) and its reconstruction by the frozen CAE ($\text{SPE}_t = 96.7$, against a D3 mean of 86).

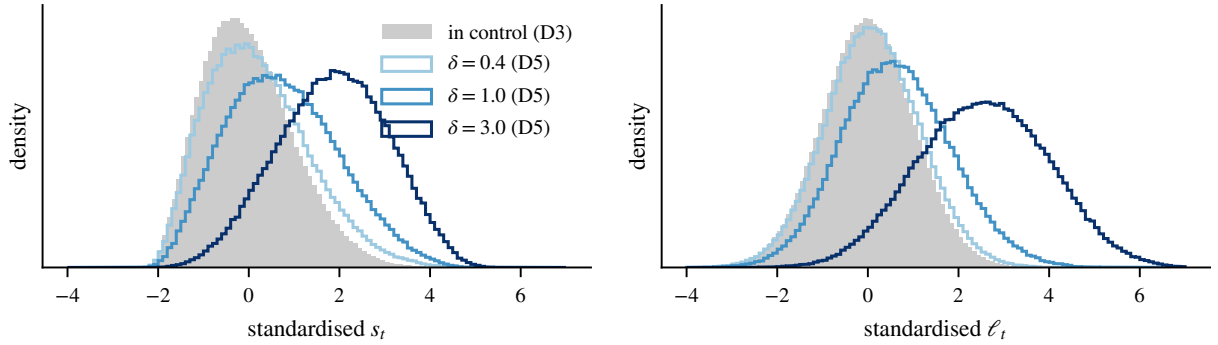


Figure 5: Post-hoc descriptive diagnostic: distributions of s_t and ℓ_t , standardised by the D3 moments, in control (D3) and at $\delta = 0.4, 1.0$ and 3.0 (first 100 D5 trajectories each). Mean shifts: s_t 0.21, 0.75, 1.86 and ℓ_t 0.10, 0.64, 2.55 standard deviations.

C Supplementary Phase II results

Figure 6 is the repository’s own summary of the D4 validation in Table 1. Tables 4–6 give the complete out-of-control results; every entry is computed from the committed per-trajectory run lengths, and no D5 run length was censored. Figure 7 shows run-length distributions of the primary pair at four shifts.

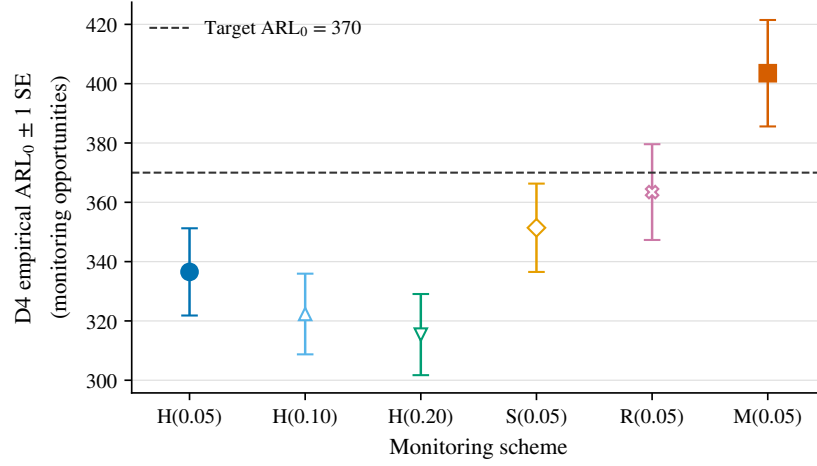


Figure 6: D4 in-control ARL (± 1 SE, 500 trajectories) of the six charts against the target of 370 (repository figure).

Table 4: Restricted out-of-control ARL (SE) of all six charts across the shift grid (D5, 500 trajectories per shift).

δ	$H(0.05)$	$H(0.10)$	$H(0.20)$	$S(0.05)$	$R(0.05)$	$M(0.05)$
0.2	282.4 (12.0)	278.1 (11.9)	283.6 (12.9)	292.1 (13.4)	336.6 (14.7)	285.6 (12.7)
0.4	229.4 (9.8)	226.3 (9.9)	220.5 (9.9)	235.7 (9.9)	298.4 (13.8)	131.1 (6.0)
0.6	165.3 (6.8)	160.5 (7.0)	155.8 (7.1)	185.8 (8.1)	220.7 (11.3)	59.6 (2.9)
0.8	110.0 (4.9)	105.8 (4.9)	99.0 (4.9)	130.4 (5.4)	160.5 (6.6)	27.7 (1.6)
1.0	78.8 (3.3)	77.7 (3.4)	76.2 (3.6)	103.8 (4.4)	126.1 (5.6)	12.4 (0.9)
1.2	56.1 (2.4)	55.8 (2.6)	53.5 (2.6)	88.6 (3.7)	90.1 (3.9)	7.3 (0.5)
1.4	42.0 (1.8)	40.8 (1.9)	36.4 (1.8)	64.7 (2.9)	69.4 (2.9)	3.1 (0.2)
1.6	30.9 (1.2)	28.4 (1.3)	25.6 (1.3)	61.8 (2.8)	53.0 (2.3)	1.7 (0.1)
1.8	26.1 (1.0)	24.1 (1.1)	22.2 (1.1)	47.7 (2.1)	45.6 (2.0)	1.2 (0.0)
2.0	19.8 (0.7)	17.9 (0.7)	15.2 (0.8)	47.3 (2.0)	35.1 (1.7)	1.1 (0.0)
2.2	16.9 (0.7)	14.9 (0.7)	12.5 (0.7)	36.5 (1.7)	31.7 (1.4)	1.0 (0.0)
2.4	14.7 (0.5)	13.0 (0.5)	10.1 (0.5)	39.1 (1.7)	23.9 (1.1)	1.0 (0.0)
2.6	12.5 (0.4)	10.6 (0.4)	8.1 (0.4)	34.8 (1.7)	21.8 (1.0)	1.0 (0.0)
2.8	12.8 (0.5)	10.9 (0.5)	8.4 (0.5)	32.7 (1.5)	21.9 (1.1)	1.0 (0.0)
3.0	12.7 (0.5)	11.2 (0.5)	9.2 (0.5)	33.1 (1.4)	21.3 (1.0)	1.0 (0.0)

Table 5: Primary comparison in detail: paired mean difference $\bar{d}$ (SE), median run lengths, share of $M(0.05)$ trajectories signalling at the first opportunity (no $H(0.05)$ trajectory did at any shift), mean delay in post-change observations ($ARL_1 + 31$) and the ratio of the two charts' delays in both units.

δ	$\bar{d}$ (SE)	MRL		$P(RL_M = 1)$	Obs. delay		Ratio H/M	
		H	M		H	M	obs.	opp.
0.2	-3.2 (15.9)	205.0	190.0	0.024	313.4	316.6	0.99	1.0
0.4	98.2 (9.7)	164.0	89.5	0.068	260.4	162.1	1.61	1.7
0.6	105.6 (6.7)	117.0	41.5	0.122	196.3	90.6	2.17	2.8
0.8	82.3 (4.8)	79.0	16.0	0.242	141.0	58.7	2.40	4.0
1.0	66.4 (3.2)	54.5	4.5	0.408	109.8	43.4	2.53	6.4
1.2	48.8 (2.3)	37.5	1.0	0.572	87.1	38.3	2.27	7.6
1.4	38.9 (1.7)	29.0	1.0	0.744	73.0	34.1	2.14	13.5
1.6	29.2 (1.2)	21.0	1.0	0.902	61.9	32.7	1.89	18.3
1.8	24.9 (1.0)	18.0	1.0	0.944	57.1	32.2	1.77	21.5
2.0	18.7 (0.7)	14.0	1.0	0.982	50.8	32.1	1.58	17.7
2.2	15.9 (0.7)	12.0	1.0	0.998	47.9	32.0	1.50	16.7
2.4	13.7 (0.5)	11.0	1.0	0.998	45.7	32.0	1.43	14.7
2.6	11.5 (0.4)	10.0	1.0	0.998	43.5	32.0	1.36	12.4
2.8	11.8 (0.5)	9.0	1.0	1.000	43.8	32.0	1.37	12.8
3.0	11.7 (0.5)	9.0	1.0	1.000	43.7	32.0	1.37	12.7

Table 6: Paired mean run-length differences (SE) of $H(0.05)$ against its ablations; negative values mean that $H(0.05)$ signalled sooner.

δ	$RL_H - RL_S$	$RL_H - RL_R$
0.2	-9.7 (14.5)	-54.1 (15.5)
0.4	-6.4 (11.3)	-69.0 (14.2)
0.6	-20.6 (8.1)	-55.4 (10.6)
0.8	-20.4 (5.3)	-50.5 (6.6)
1.0	-25.0 (4.2)	-47.3 (5.1)
1.2	-32.5 (3.5)	-33.9 (3.7)
1.4	-22.7 (2.8)	-27.4 (2.7)
1.6	-30.9 (2.7)	-22.1 (2.2)
1.8	-21.6 (1.9)	-19.5 (1.8)
2.0	-27.5 (2.0)	-15.3 (1.6)
2.2	-19.6 (1.6)	-14.8 (1.3)
2.4	-24.3 (1.7)	-9.1 (1.0)
2.6	-22.2 (1.6)	-9.2 (0.9)
2.8	-19.8 (1.4)	-9.0 (1.0)
3.0	-20.4 (1.4)	-8.6 (0.9)

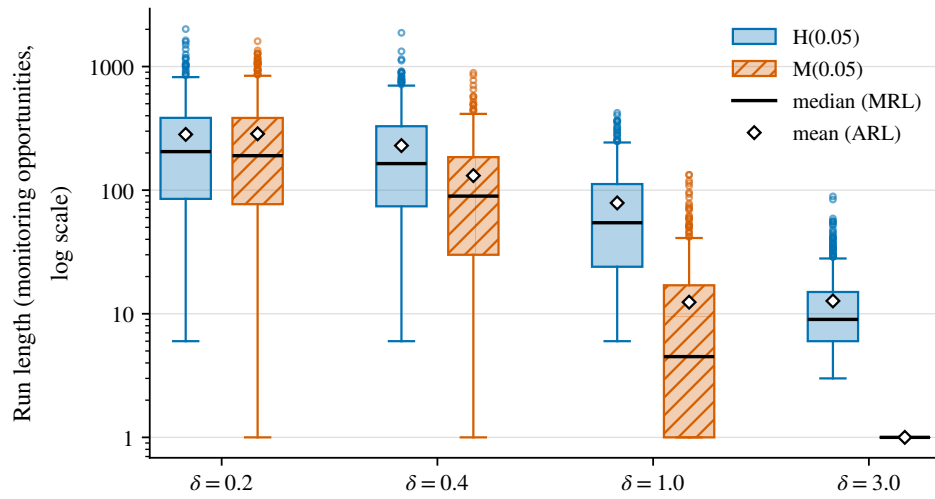


Figure 7: Run-length distributions of $H(0.05)$ and $M(0.05)$ at four shifts (log scale; repository figure).

D Reproducibility and provenance

The experiment is the committed state of the authoritative repository `pipeline-thesis-independent` (branch `simplify-pipeline-code`, commit `e62bdee`). Data generation uses master seed 795 with independent PCG64DXSM streams per dataset, trajectory and shift; the CAE training seed is 3795; all numerical routes run in double precision (network in single precision) on one CPU thread. The frozen preprocessing statistics, CAE weights, OC-SVM state, D3 moments and control limits, and the per-trajectory D4/D5 run lengths are committed artifacts, and the production run that produced them took 207 s. Every figure and generated table in this report is rebuilt, without refitting or recalibration, by `scripts/make_all_figures.py` in the report workspace, which reads these artifacts read-only; the source and derivation of each figure are recorded in `provenance/final_figure_provenance.md`. The post-hoc diagnostic audit cited in Section 4.5 is a separate analysis of the same frozen experiment and is not part of its pre-specified design.